

Lesson 2 · Exponentials and Logarithms

The one idea: a log is a power

$\log_b y$ asks one question: **what power of b gives y ?**

$$2^3 = 8 \iff \log_2 8 = 3 \quad y = b^x \iff \log_b y = x$$

Every question in this lesson is a switch between these two forms. Say it as you write it: **base to the bottom, y in the brackets.**

1 What a log means

Power form	Log form	Value
$3^4 = 81$	$\log_3 81 = 4$	4
$10^{-2} = 0.01$	$\log 0.01 = -2$	-2
$9^{\frac{1}{2}} = 3$	$\log_9 3 = \frac{1}{2}$	$\frac{1}{2}$
$b^0 = 1$	$\log_b 1 = 0$	0
$b^1 = b$	$\log_b b = 1$	1

- log with no base written means $\log_{10}$: the calculator's **log** key.
- $\ln$ means $\log_e$, where $e \approx 2.718$: the calculator's **ln** key. e^x and $\ln x$ undo each other.
- The base is positive and not 1. You can only take the log of a **positive** number.

Write it in this order (an equation with one log)

1. Get the log on its own: $\log_b(\text{something}) = k$.
2. Switch to power form: $\text{something} = b^k$.
3. Solve. If the unknown is the base, the base must be positive: $\log_x 49 = 2$ gives $x^2 = 49$, so $x = 7$, not -7 .

Example 1. Solve $\log_5 x = 3$. $x = 5^3 = 125$

(u)

2 The log laws

These are on the formula sheet. Each one is an index law in disguise.

Log law	Index law behind it	Example
$\log_b(xy) = \log_b x + \log_b y$	multiply: add the powers	$\log_2 3 + \log_2 5 = \log_2 15$
$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$	divide: subtract the powers	$\log_2 12 - \log_2 4 = \log_2 3$
$\log_b(x^n) = n \log_b x$	power of a power: multiply	$\log x^3 = 3 \log x$

There is no law for $\log(x + y)$, and $\frac{\log x}{\log y}$ is **not $\log \frac{x}{y}$.**

Write it in this order (a single log)

1. Move every number in front of a log up as a power: $2 \log x = \log x^2$.
2. Plus signs multiply inside one log; minus signs divide.
3. Simplify the fraction inside.

Example 2. Write $2 \log x + \log 5 - \log 10$ as a single log.

$$= \log x^2 + \log 5 - \log 10 = \log \frac{5x^2}{10} = \log \frac{x^2}{2}$$

Example 3. Solve $\log_3(x+4) + \log_3(x-4) = 2$.

$$\begin{aligned}\log_3((x+4)(x-4)) &= 2 \\ x^2 - 16 &= 3^2 = 9 \\ x &= 5 \quad (x = -5 \text{ rejected})\end{aligned}$$

(r)

(u) $x = -5$ makes $x - 4$ negative, and the log of a negative number does not exist.

3 Solving exponential equations

Write it in this order (unknown in the power)

1. Get the power on its own: $b^{(\dots)} = k$.
2. Same base on both sides? Match the powers and solve exactly. Otherwise:
3. Take logs of both sides and bring the power down **in a bracket**: $(2x - 1) \log 5 = \log 200$.
4. Divide by the log, then finish the rearranging. Round only at the end.
5. For e , use $\ln$: $e^{0.3t} = 5$ gives $0.3t = \ln 5$.

Example 4. Solve $5^{2x-1} = 200$.

$$\begin{aligned}(2x - 1) \log 5 &= \log 200 \\ 2x - 1 &= \frac{\log 200}{\log 5} = 3.2920 \dots \\ x &= 2.146\end{aligned}$$

(u)

Example 5. Solve $4^{x+1} = 8^x$ exactly.

$$2^{2(x+1)} = 2^{3x} \Rightarrow 2x + 2 = 3x \Rightarrow x = 2$$

(u)

Example 6. Solve $3^{2x} - 4 \times 3^x + 3 = 0$.

Let $u = 3^x$, so $3^{2x} = u^2$:

$$\begin{aligned}u^2 - 4u + 3 &= 0 \\ (u - 1)(u - 3) &= 0 \\ 3^x &= 1 \text{ or } 3^x = 3 \\ x &= 0 \text{ or } x = 1\end{aligned}$$

(t)

4 Growth and decay

$P = Ar^t$ A = starting amount, r = growth factor per time step, t = number of steps

- Growth of $R\%$: $r = 1 + \frac{R}{100}$ (6% growth: $r = 1.06$). Decay of $R\%$: $r = 1 - \frac{R}{100}$ (15% loss: $r = 0.85$).
- **When** questions: put the numbers in, divide by A , take logs. **What rate** questions: divide, then take the root (t -th root).
- Finish with a **sentence in context**, with units, and say what the decimal means ("during the 7th year").

Example 7. A car bought for \$18 000 loses 15% of its value each year. When is it first worth less than \$6000?

$$\begin{aligned}18\,000(0.85)^t &= 6000 \\ 0.85^t &= \frac{1}{3} \\ t &= \frac{\log \frac{1}{3}}{\log 0.85} = 6.76\end{aligned}$$

(r)

It is first worth less than \$6000 **during the 7th year**, after about 6.8 years.

(r)

Example 8. \$5000 grows to \$7500 in 8 years at a fixed annual rate $R\%$. Find R .

$$\begin{aligned}5000\left(1 + \frac{R}{100}\right)^8 &= 7500 \\ 1 + \frac{R}{100} &= 1.5^{\frac{1}{8}} = 1.0520 \\ R &= 5.2\%\end{aligned}$$

5 Versions the marker won't accept

Question	Answer given	Why it falls short
Show that $D = \frac{\log 2}{\log(1 + \frac{R}{100})}$	the final line only	A "show that" needs the line where logs are taken of both sides: markers look for that line.
Solve $\log_x 49 = 2$	$x = \pm 7$	A base cannot be negative. The answer is $x = 7$ with -7 rejected for that reason.
How long until ...?	$t = 6.7599$	No units, no meaning. Give a sentence: "during the 7th year".
Write as a single logarithm	$\log 3a + \log \frac{a^2}{36}$	Two logs are not a single log, and the fraction inside is not simplified.

6 Ways to lose marks

From the wording of the marking schedules and from teaching observation; numbered for marking.

- L11 Exponential and log forms mixed up.** $\log_2 32 = 5$ means $2^5 = 32$. Base to the bottom, y in the brackets.
- L12 Log laws used on a sum, or on a quotient of logs.** $\log(x + 3) \neq \log x + \log 3$, and $\frac{\log 40}{\log 5} \neq \log 8$.
- L13 Power not bracketed when it comes down.** $\log 5^{x-2} = (x - 2) \log 5$, not $x - 2 \log 5$.
- L14 Log of zero or a negative number, or a negative base, kept.** Check every answer in the original logs: each bracket must be positive, and a base must be positive.
- L15 Growth factor wrong.** A 20% loss each year multiplies by 0.8, not 0.2.
- L16 Answer not interpreted in context.** Units, a sentence, and rounding only at the end.

7 Quick review

Powers to know by sight	Index laws behind the log laws
$2^5 = 32$ $2^6 = 64$ $2^7 = 128$ $2^{10} = 1024$	$x^a \times x^b = x^{a+b}$ $\log(xy) = \log x + \log y$
$3^4 = 81$ $3^5 = 243$ $5^3 = 125$ $5^4 = 625$	$x^a \div x^b = x^{a-b}$ $\log \frac{x}{y} = \log x - \log y$
$4^3 = 64$ $6^3 = 216$ $7^3 = 343$ $10^{-2} = 0.01$	$(x^a)^n = x^{an}$ $\log x^n = n \log x$